INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April – May, 2025

Degree: **BCA** Semester: **Second**

Stream: **BCA**

Paper Code: **BCACC203**

Paper Name: **Computer Architecture**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1. What would be the decimal equivalent of the binary number 111.111? **2**

2. What would be the 2's complement representation of the decimal value 15? **2**

3. Which base 7 number is equivalent to the decimal number 179? **2**

4. Find the hexadecimal number which is equivalent to the decimal number 101.125? **2**

5. 10111 is the 2's complement representation of which decimal number? **2**

6. What is the 9's complement of $(0.3267)_{10}$? **2**

7. In which addressing modes, the operand value is directly specified? **2**

8. In order to realize an adder that can add two 16-bit numbers, how many full-adders and half-adders would be required? **2**

9. The IR stores which type of instructions? **2**

10. In which addressing modes is the actual operand provided as a part of the instruction? **2**

11. In a two-pass assembler, what does the first pass achieve? **2**

12. "The last instruction written by the programmer as the interrupt service routine must restore the PC with the address of the last executed program instruction." – Is this statement TRUE regarding the interrupt mechanism? Explain. **2**

13. The contents of the multiplicand register and the multiplier register of a hardware circuit implementing Booth's algorithm are the binary numbers 111001 and 111100. What will be the result produced by the multiplier circuitry (in decimal)? **2**

14. In the implementation of a binary multiplier circuit, which set of logic gates is used? **2**

15. Divide overflow is easiest to handle in which circuit implementing division? **2**

Part - B

Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Convert the following decimal numbers to the bases indicated. **5**
    a. 7562 to octal
    b. 1938 to hexadecimal
    c. 175 to binary

17. Perform the arithmetic operations (+70) + (+80) and (-70) + (-80) with binary numbers in signed -2's complement representation. Use eight bits to accommodate each number together with its sign. Show that overflow occurs in both cases, that the last two carries are unequal, and that there is a sign reversal. **5**

18. Represent the number $(+46.5)_{10}$ as a floating-point binary number with 24 bits. The normalized fraction mantissa has 16 bits and the exponent has 8 bits. **5**

19. Register A holds the 8-bit binary 11011001. Determine the B operand and the logic microoperation to be performed in order to change the value in *A* to: **5**
    a. 01101101
    b. 11111101

20. Starting from an initial value of $R = 11011101$, determine the sequence of binary values in *R* after a logical shift-left, followed by a circular shift-right, followed by a logical shift-right and a circular shift-left. **5**

21. A computer uses a memory unit with 256K words of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts: an indirect bit, an operation code, a register code part to specify one of 64 registers, and an address part. **5**

a) How many bits are there in the operation code, the register code part, and the address part?
b) Draw the instruction word format and indicate the number of bits in each part.
c) How many bits are there in the data and address inputs of the memory?

22. A non-pipeline system takes 50 ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 10 ns. Determine the speedup ratio of the pipeline for 100 tasks. What is the maximum speedup that can be achieved? **5**

23. Convert the following numerical arithmetic expression into reverse polish notation and show the stack operations for evaluating the numerical result. **5**
$(3 + 4)[10(2 + 6) + 8]$

24. A computer has 32-bit instructions and 12-bit addresses. If there are 250 two-address instructions, how many one-address instructions can be formulated? **5**

Part - C
Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. a. perform the operation (-9) + (-6) = -15 with binary numbers in signed -1's complement representation using only five bits to represent each number (including the sign). Show that the overflow detection procedure of checking the inequality of the last two carries fails in this case. **10**

b. Suggest a modified procedure for detecting an overflow when signed -1's complement numbers are used.

26. Design an array multiplier that multiplies two 4-bit numbers. Use AND gates and binary adders. **10**

27. Present a flowchart for Booth algorithm for multiplication of signed -2's complement numbers and describe it. **10**

28. a. How many 128 × 8 RAM chips are needed to provide a memory capacity of 2048 bytes? **10**
b. How many lines of the address bus must be used to access 2048 bytes of memory? How many of these lines will be common to all chips?
c. How many lines must be decoded for chip select? Specify the size of the decoders.

29. Convert the following arithmetic expressions from reverse polish notation to infix notation. **10**
a. $A\ B\ C\ D\ E + * - /$

b. $A\ B\ C\ D\ E\ */-+$
c. $A\ B\ C\ */D-E\ F/+$
d. $A\ B\ C\ D\ E\ F\ G+*+*+*$

30. Convert the following arithmetic expressions from infix to reverse polish notation. **10**

a. $A * B + C * D + E * F$
b. $A * B + A\ C\ (B * D + C * E)$
c. $A + B * [C * D + E * (\mathrm{F} + \mathrm{G})]$
d. $\frac{A * [B + C * (D + E)]}{E * (\mathrm{F} + \mathrm{G})}$

31. Write a program to evaluate the arithmetic statement: **10**

$$\frac{A * (B * (C+D) - E) + F}{G * (\mathrm{H} + 1) - K}$$

32. What are the major characteristics of a RISC processor? Briefly describe the difference between a RISC and CISC processor? **10**

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